

Multimodal Applications

Generative AI

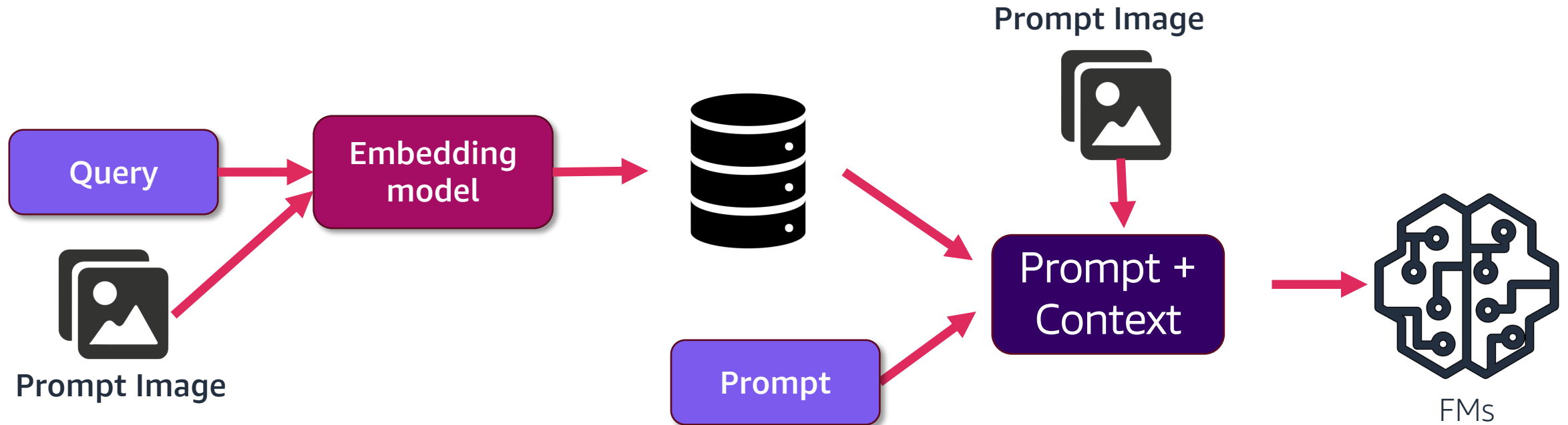
Module 3 – Lesson 5

Today's activities

- Review
- Examples of multimodal applications
- Multimodal agents

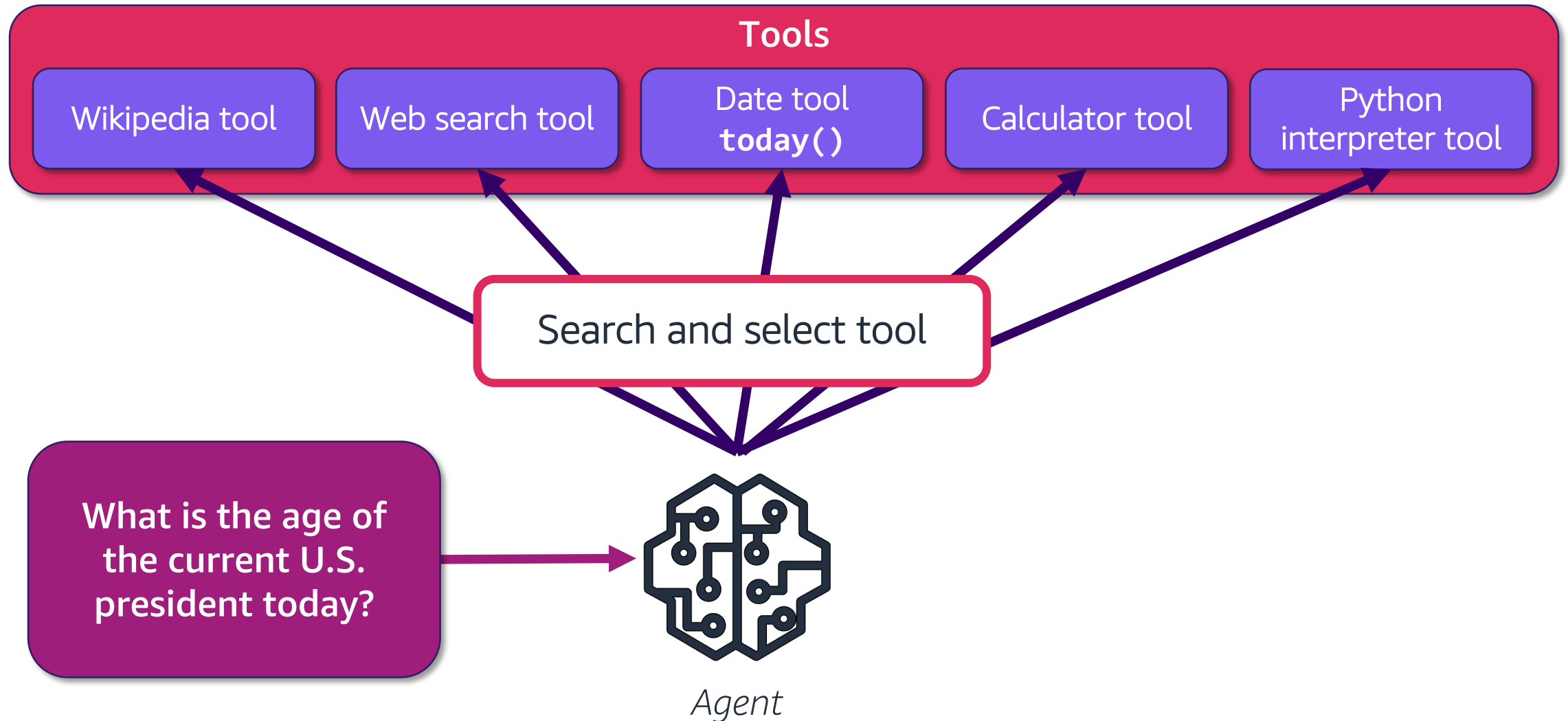


Review: Multimodal RAG workflow



- Use the same embedding model for the vector database and embedding the query
- Images can be used for retrieval, prompting and both

Review: Agents with tools



How do multimodal applications add value to businesses?

- **Improved accuracy and robustness**
 - Richer representation from multiple modalities
- **Ease of development**
 - Complex models but simpler applications
- **Interactive and intuitive solutions**
 - Interact using natural mediums such as natural language text, speech, gestures, etc.
- **Easy access to highly performant open-source and commercial models**
 - Only pay to use, not to pre-train
 - Wide range of model choices (size, cost, performance, etc.)

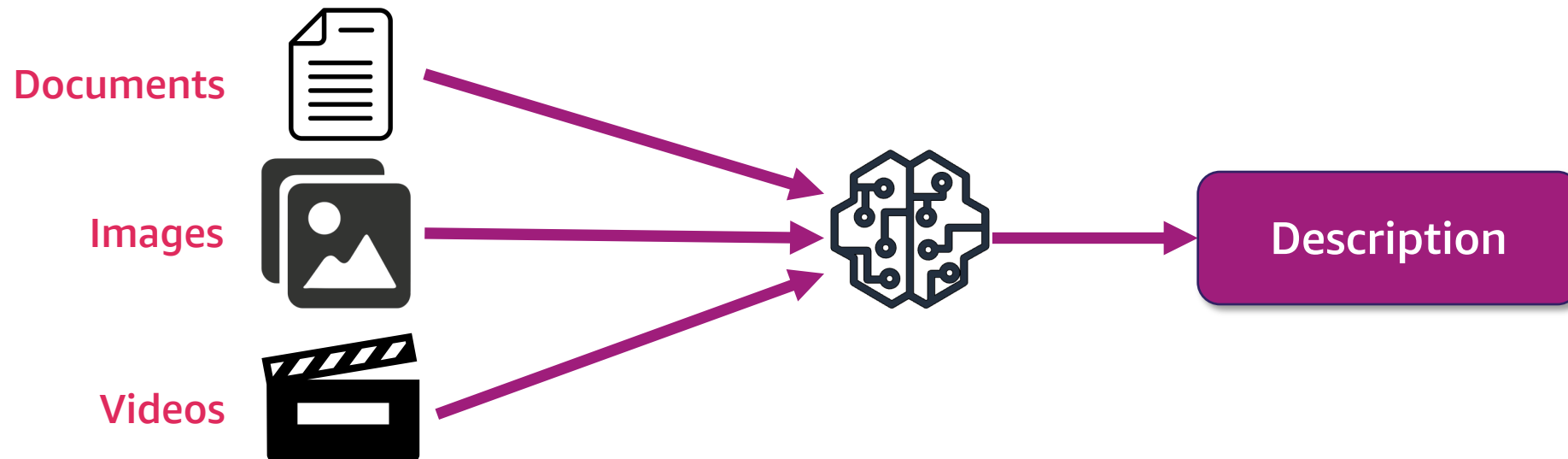
Examples of multimodal applications

Personalization

- Automatically generate or update content for **target audience**
 - Match the expected technical depth
 - Provide relevant examples and anecdotes to explain concepts
 - Define length of documents/videos based on requirements
 - Use of inclusive and accessible language
 - Multilingual content generation
- **Personalized content consumption**
 - Interact with documents/presentations/videos
 - Q&A for summaries, detailed explanations, translations, scene descriptions, etc.
 - Personalized self-paced learning

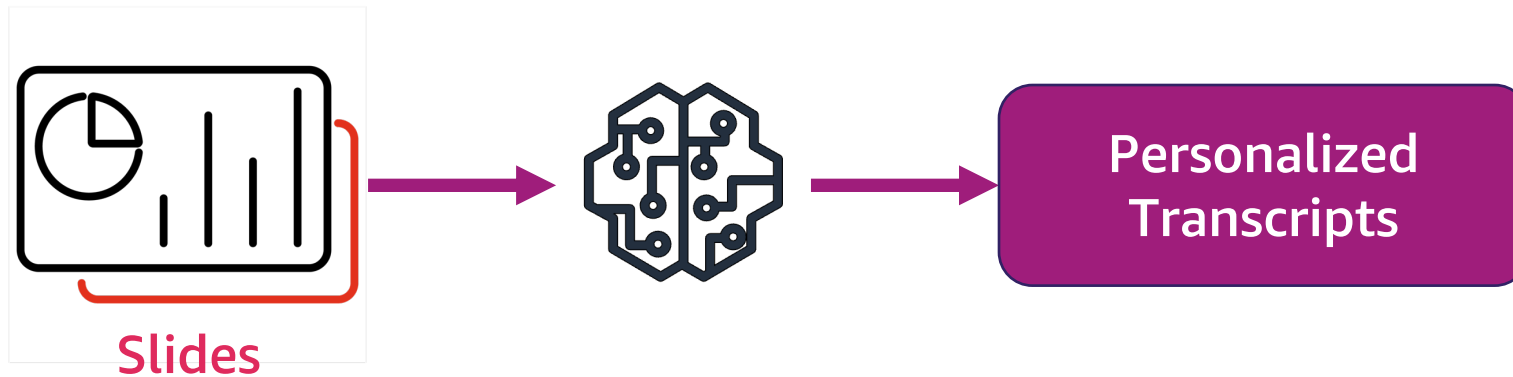
Personalization: Scene description

- Descriptions of visual elements
- Alt-text generation
- Multilingual descriptions



Personalization: Personalized presentations example

- Generate **personalized transcripts** and notes
 - Eg: Academic courses tailored for each student/ group
- Examples and anecdotes that appeal to audience's interest
- **Minimal human supervision**
 - Proofing required



Video Analysis (1/3)

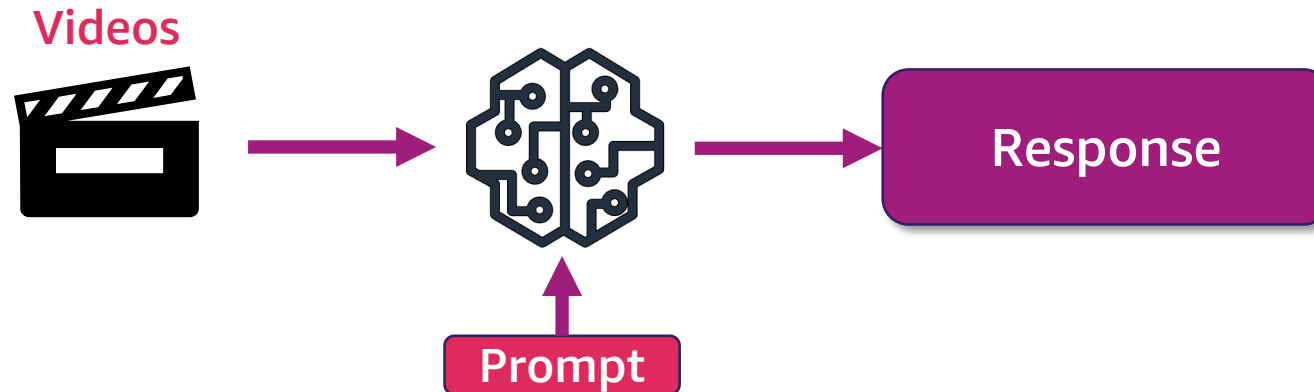
- **Navigate videos**
 - Jump to exactly when a topic was first introduced
 - Find all mentions of a concept
 - Skip to next topic
- **Summarize videos**
 - Summarize entire videos
 - Summarize explanation of each concept
 - Summarize within time range (eg: first 10 mins)

Video Analysis (2/3)

- Personalize video consumption
 - Detailed explanations of concepts or short summaries
 - Provide examples to support concepts presented in videos
- Multilingual Q&A for improved accessibility
- Extract step-by-step instructions from walkthrough videos
 - Troubleshoot and assist with setup process

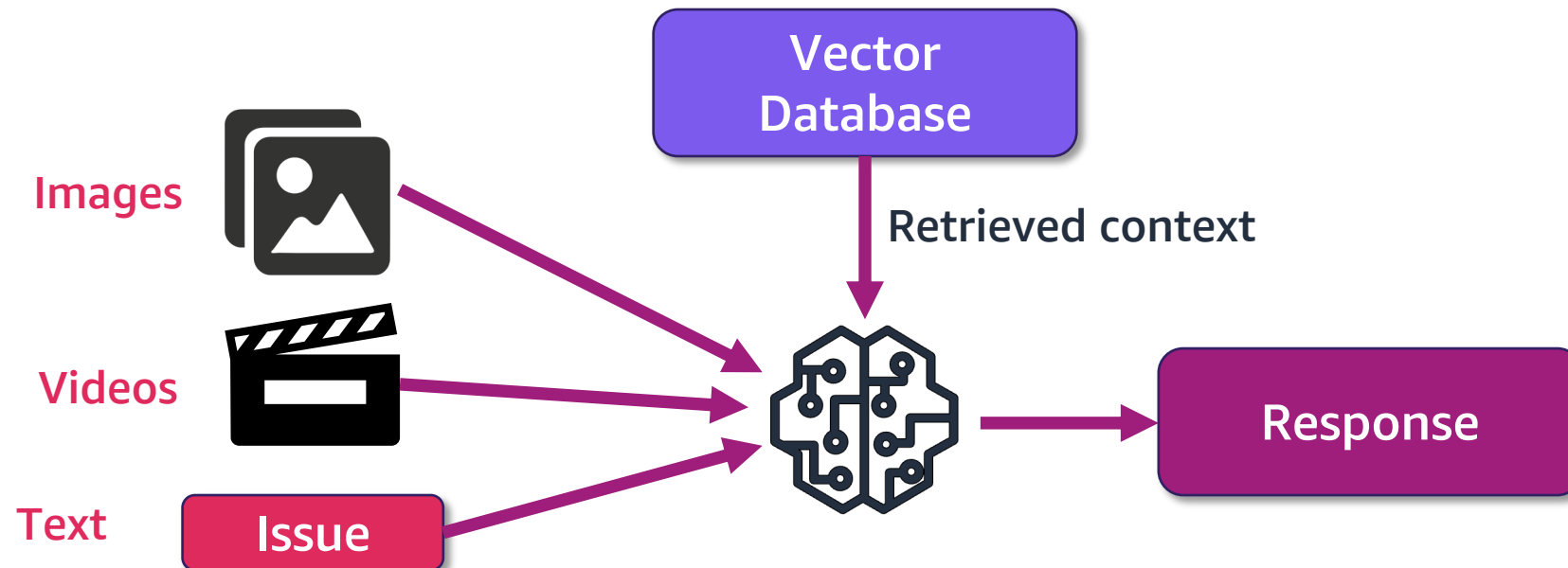
Video Analysis (3/3)

- Directly chat with videos
- Model capable of video understanding
- All the information is retrieved directly from the video
 - Challenging for some models to extract all the details with high recall



Customer service applications (1/2)

- Describe the issue with text, images, videos, etc.
- Vector database contains relevant documents for resolution
 - User manual, FAQs, troubleshooting guide, etc.



Customer service applications (2/2)

- Identify and resolve customer complaints
 - Using textual feedback
 - Using images/videos uploaded by customer
- Automated call support using multimodal solutions
 - Identify urgency, frustration, etc., from audio and determine if escalation is appropriate
 - Guided assistance via video calls
- Q&A with videos to resolve issues
 - Tutorial videos
 - Discussed in the previous section

E-commerce

- Multimodal search engines
- Find similar products using images, text and videos
- Generate product listing from images/videos
- Automatically extract product information from image/label
- Generate images of products in different scenes

Use case: Healthcare

- Generate preliminary reports from scans based on patient records
- Assistance with triaging patients
- Virtual diagnosis of minor ailments
- Generate personalized reports for patients

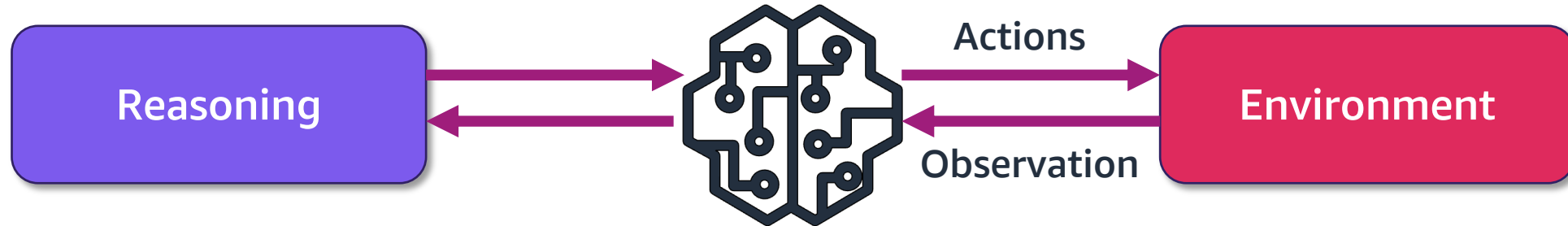
Use case: Accessibility

- Scene descriptions from images/videos
- ASL content for hearing challenged
- Multimodal screen readers
- Fall/hurt detection for the elderly
- Personalized content generation
 - Improved inclusivity
 - Cater to different demographics
- Analysis of multilingual content

Multimodal Agents

Review: ReAct

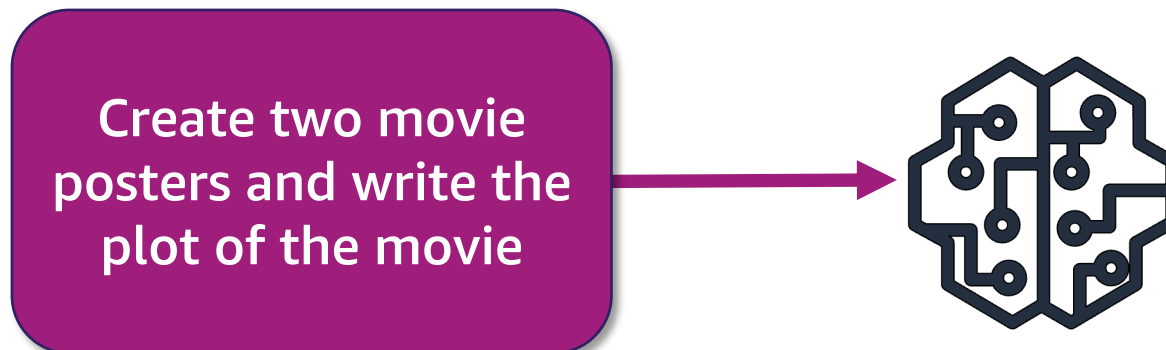
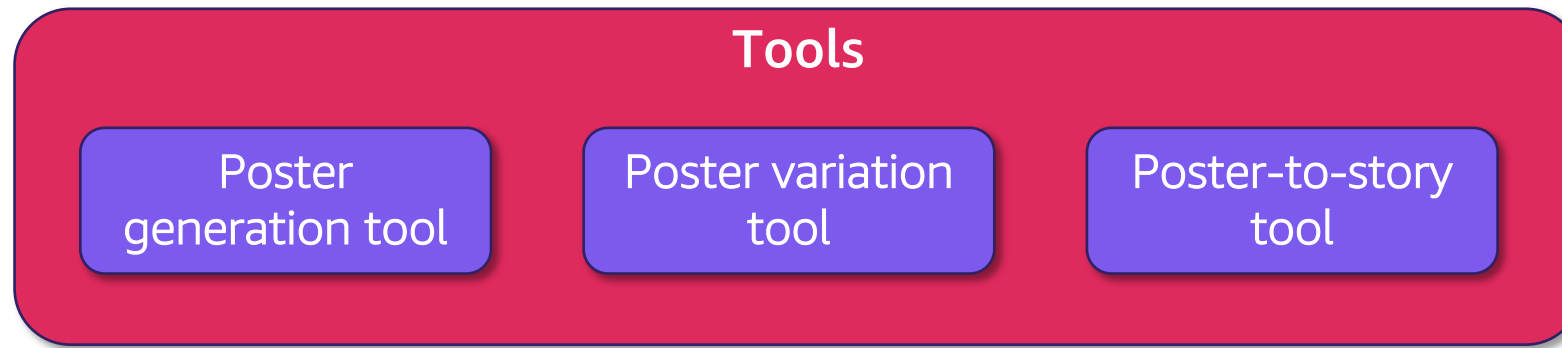
- LLMs often demonstrate the ability to **reason** logically.
- LLMs are capable of generating **action** plans.
- Combine both!
 - **Reasoning**: Create, track, and update action plans. Handle errors.
 - **Acting**: Interface with functions, tools, knowledge bases, or environments.



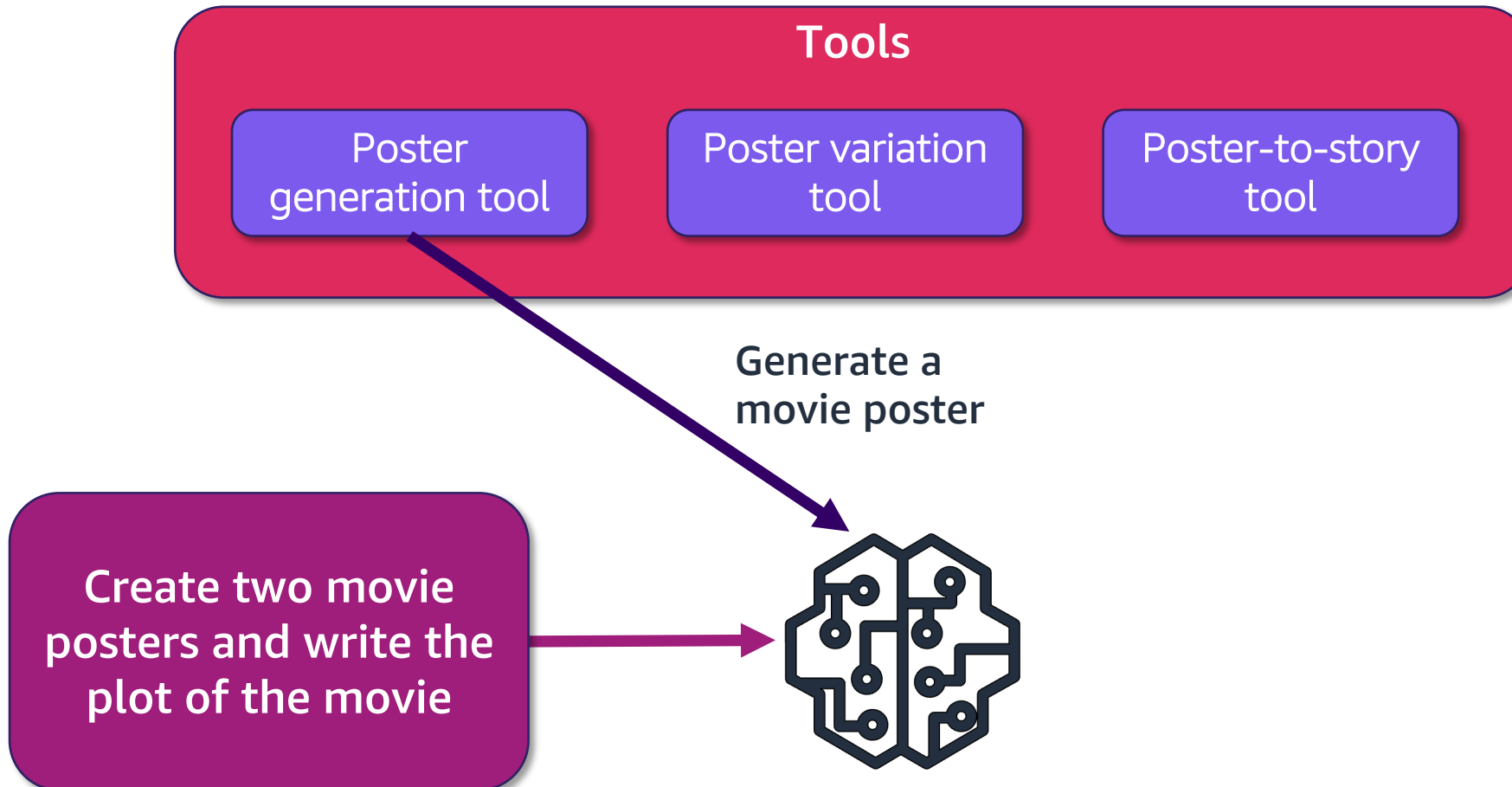
Multimodal agents' implementation

- **LangChain agents** can use ReAct to plan and execute a task.
- Agents may have **access to tools and APIs**.
 - Tools need to be described so that the agent is aware of its function.
 - Important to give the agent access to right tools for a task.
- Currently, LangChain agents only support text
 - We will use custom tools to build multimodal agents.

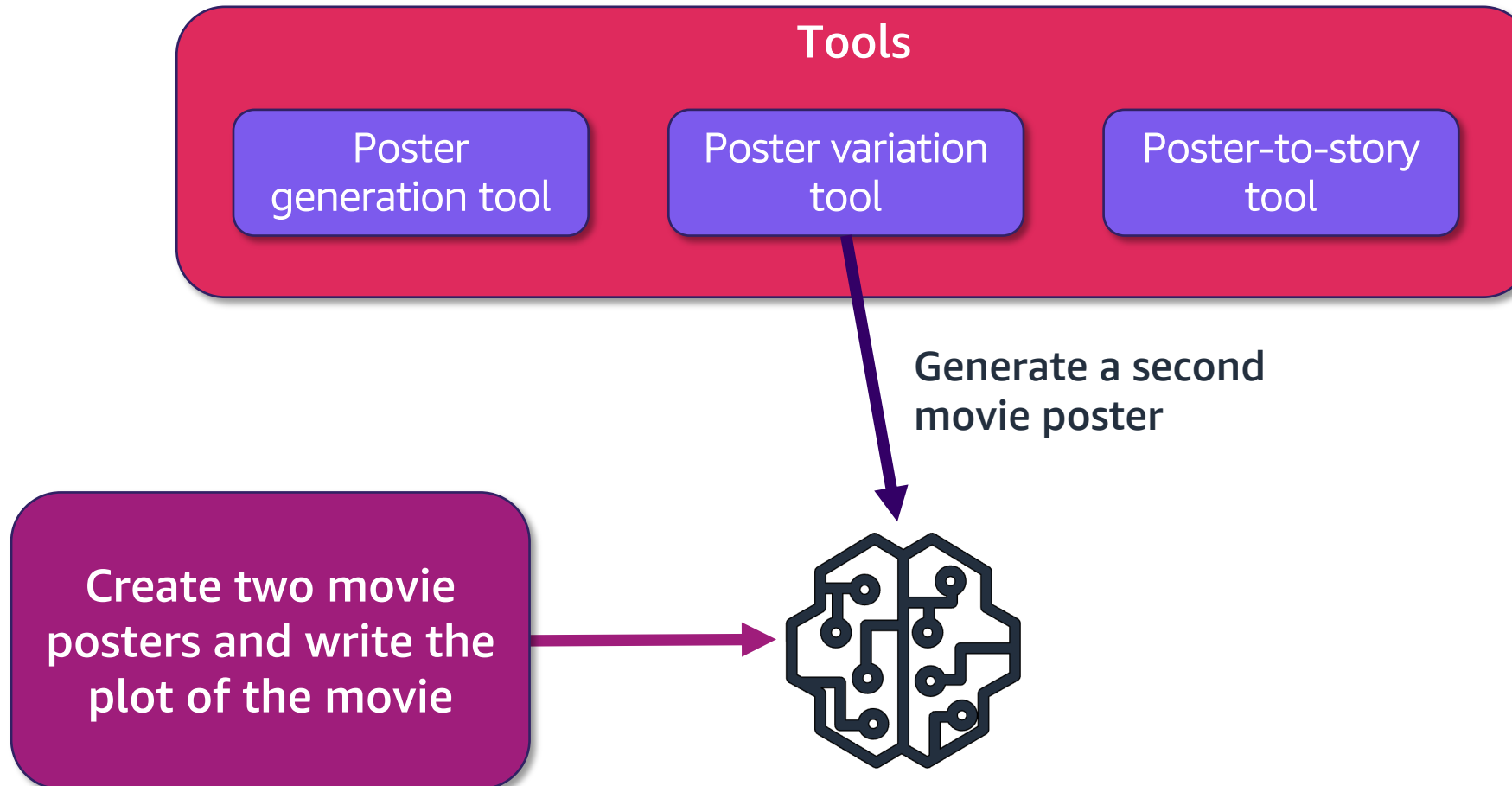
Example 1: Image generation and description (1/4)



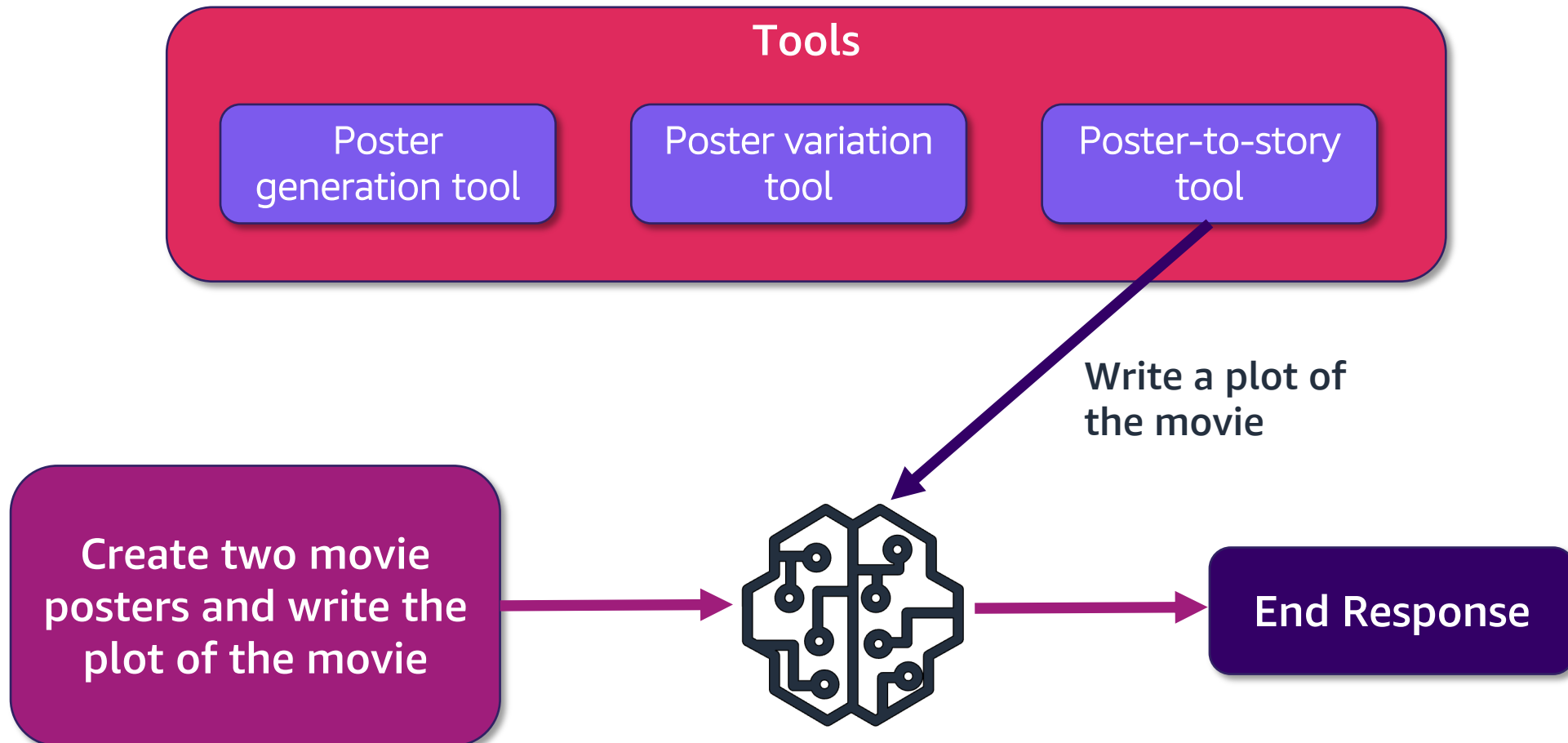
Example 1: Image generation and description (2/4)



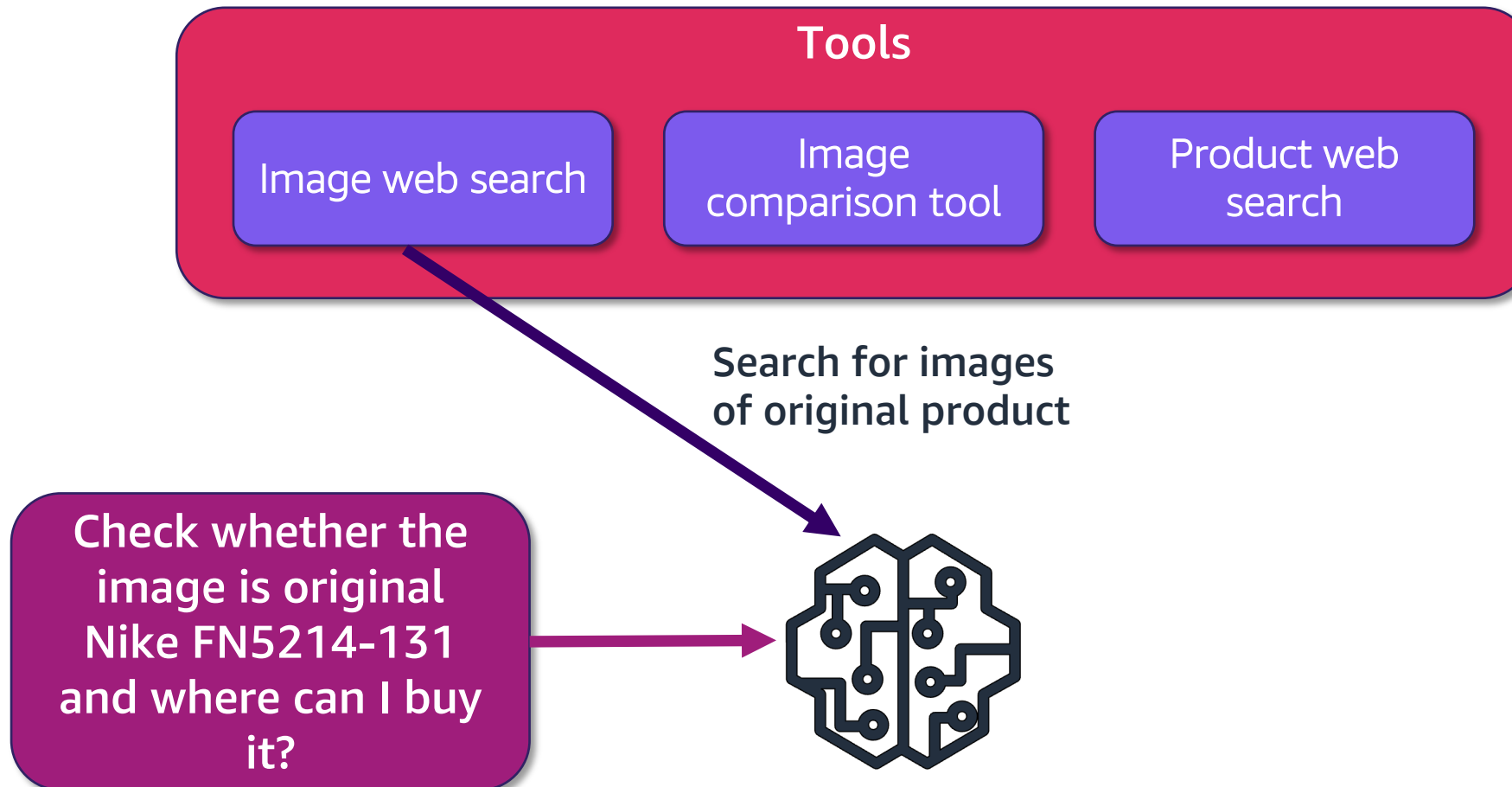
Example 1: Image generation and description (3/4)



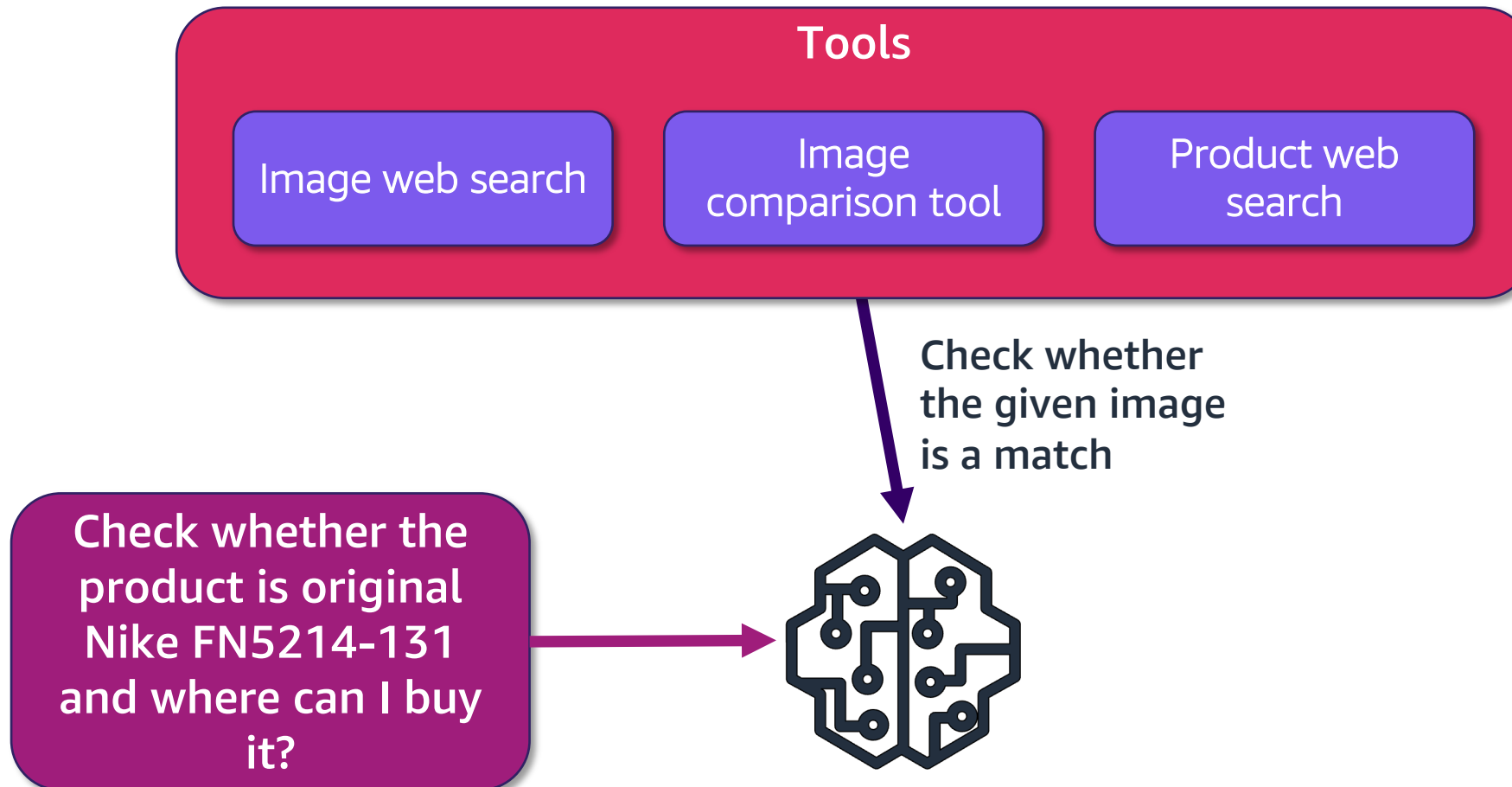
Example 1: Image generation and description (4/4)



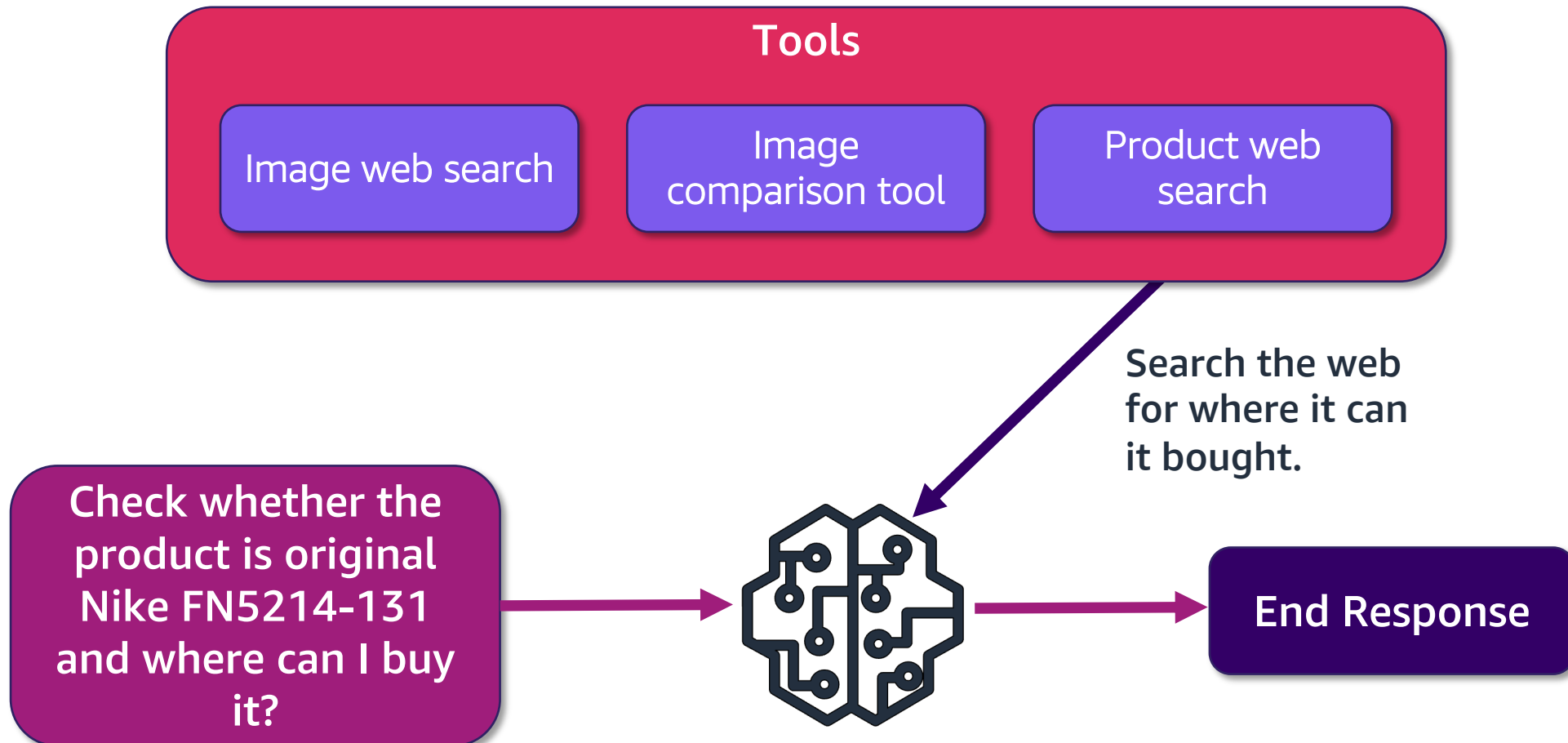
Example 2: Multimodal agent for retrieval-based responses (1/3)



Example 2: Multimodal agent for retrieval-based responses (2/3)

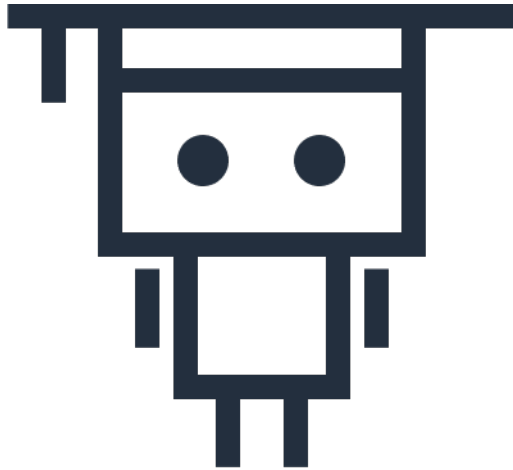


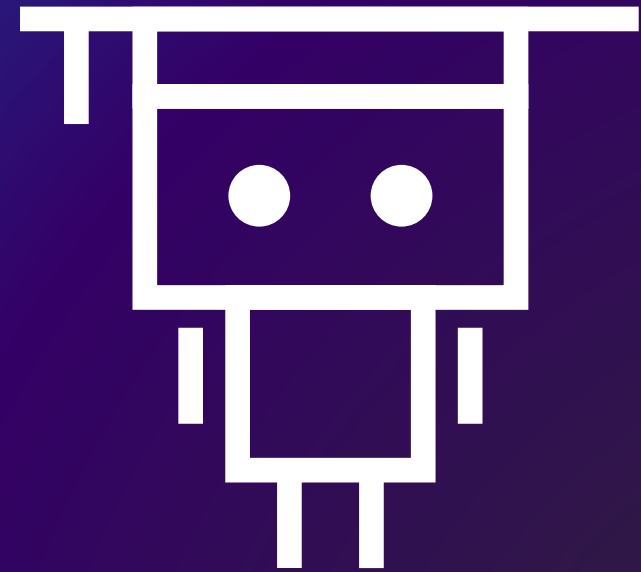
Example 2: Multimodal agent for retrieval-based responses (3/3)



Next lesson

- This lesson covered topics related to multimodal applications





Thank you!